Roshan Mishra

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PROFILE SUMMARY

AI Engineer with 3 years of experience building and deploying **LLM, RAG, and agentic AI systems** for enterprise clients. Built an enterprise document assistant that raised retrieval precision and recall on keyword searches from **under 70% to over 85%**, and an AI workflow that cut a **29-minute compliance task to about 7 seconds**. Works across the full lifecycle, from data ingestion and retrieval to evaluation and AWS deployment.

KEY SKILLS

AI & ML: LLMs, RAG, Hybrid Retrieval (BM25 + Embeddings), Agentic Workflows, Tool Calling, Structured Outputs, Prompt Engineering, Retrieval & LLM Evaluation, Fine-Tuning (QLoRA, LoRA, PEFT), Quantization
Deep Learning Frameworks: PyTorch, Hugging Face Transformers
Backend Engineering: Python, FastAPI, REST APIs, Pydantic, Async & Parallel Processing, Docker
Cloud: AWS (Lambda, ECS, EC2, S3, Bedrock), Azure OpenAI
LLM Tooling: Anthropic Claude, LangChain, LangGraph, Elasticsearch, Ollama
Practices: Schema Validation, Retry & Backoff Logic, Checkpointing, Logging

WORK EXPERIENCE

Data Scientist – DataEconomy Sep 2026 – Present

AI Engineer – Accenture Oct 2023 – Sep 2026

POAM Generator Automation (Agentic LLM Workflow)

- Built a production **agentic workflow on AWS Lambda + Bedrock** that generates POAM compliance justifications from CVE inputs, cutting about **29 minutes of analyst work per CVE to about 7 seconds**.
- Reduced a batch run of about 100 CVEs from **14 minutes to 2.5** with parallel workers, bounded concurrency, and caching.
- Enforced **schema-validated JSON outputs** and built targeted recovery for responses truncated at the token limit, avoiding full workflow reruns.
- Handled LLM rate limits with exponential backoff and retries; used **S3 manifests, checkpoints, and EC2 offloading** so large batches resume instead of restarting.

SAP Role Assistant (Enterprise RAG System)

- Built an enterprise **RAG assistant** answering natural-language questions over large Word, Excel, and PDF repositories, using Anthropic Claude, Azure OpenAI embeddings, and Elasticsearch; deployed with **Docker on AWS ECS**.
- Diagnosed weak retrieval on exact business terms and added **hybrid semantic + BM25 search** with normalised scores, raising keyword-query precision and recall from **under 0.70 to over 0.85** on the evaluation set.
- Designed **structure-aware chunking**: paragraph chunks for Word, tables kept row by row, and Excel chunks that repeat their headers so rows keep their meaning.
- Enforced **access control at retrieval** with metadata filters, so only chunks a user is authorised to see reach the model.
- Added relevance thresholds and an LLM judge for borderline retrievals to keep answers grounded.
- Cut ingestion time by **95%** through batching and parallel API calls, with hash-based change detection to avoid re-embedding unchanged files.

PROJECTS

Fleet Management Intent Classifier (Local LLM) — GitHub Repo

- Built a local LLM service (**Qwen 2.5 3B on Ollama, FastAPI, Pydantic**) for intent classification and slot extraction across 8 fleet-management intents, returning validated JSON.
- Added a **deterministic rule layer** around the model, raising exact-match accuracy on a 45-query held-out set from **73% to 89%** with **100% valid JSON**; shipped with Docker, CI, and 50 automated tests.

UI-Agent (Assistant-Driven Gmail Client) — GitHub Repo

- Built a Gmail web client (**Next.js, TypeScript, FastAPI**) whose assistant drives compose, reply, and filters.
- Added a **human-confirmation toggle** for assistant-driven compose, with Gmail OAuth and live inbox sync.

Low-Resource LLM Fine-Tuning & Activation Steering — GitHub Repo

- Fine-tuned a **7B parameter LLM (Mistral)** with **QLoRA (4-bit quantization + LoRA)** on a custom options-trading question set, sized to train on a single **T4 GPU (Colab)**.
- Implemented **inference-time behavior steering** on transformer hidden states with PyTorch forward hooks.

EDUCATION

B.Tech Information Technology
Vellore Institute of Technology, Vellore 2019–2023